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Bearing assembly for a gas turbine engine

Abstract

A gas turbine engine is provided that includes a static structure, a rotating shaft, and a bearing assembly. The bearing assembly includes a bearing and a seal subassembly. The bearing has inner and outer races and a plurality of roller elements. The outer race includes a shield member. The seal subassembly includes a seal element coupled with the static structure and a seal runner coupled with the rotating shaft. Portions of the seal runner, the outer race, and the inner race define a first bearing cavity. Portions of the seal runner, the outer race, the seal element, and the static structure define a lateral bearing cavity. The lateral bearing cavity is disposed axially between the first bearing cavity and the seal element. The outer race includes a plurality of drain holes that provide fluid communication between the first bearing cavity and the lateral bearing cavity.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2990202	12/1960	Dennison	277/408	F16J 15/3404
2992842	12/1960	Shevchenko	277/306	F01D 11/003
3357757	12/1966	Walton Morley	N/A	N/A
4344506	12/1981	Smith	N/A	N/A
4406460	12/1982	Slayton	277/433	F16J 15/3404
4542623	12/1984	Hovan	N/A	N/A
4928978	12/1989	Shaffer	277/401	F16J 15/162
6579010	12/2002	Trapp	384/537	F16C 35/073
7806596	12/2009	Shatz	384/493	F16C 19/548
9752616	12/2016	Saadi	N/A	F16C 33/6685
9777592	12/2016	Denis	N/A	N/A
10364846	12/2018	Clark	N/A	F16C 37/00
10753219	12/2019	Labbe	N/A	F01D 25/125
11085330	12/2020	Pankratov	N/A	F01D 25/183
11261754	12/2021	Patard	N/A	F01D 25/183
2008/0135336	12/2007	Jewess	184/6.11	F16C 33/6677
2012/0189235	12/2011	Mcneil	N/A	N/A
2016/0032840	12/2015	James	N/A	N/A
2020/0116047	12/2019	Theratil	N/A	F01D 25/183

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107060898	12/2018	CN	N/A

OTHER PUBLICATIONS

EP Search Report for EP24205048.2 dated Feb. 12, 2025. cited by applicant

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Background/Summary

BACKGROUND OF THE INVENTION

1. Technical Field

(1) The present disclosure relates to gas turbine engines in general and to a bearing assembly coupling a rotating shaft to a static structure of the turbofan gas engine in particular.

2. Background Information

(2) A gas turbine engine of a type preferably provided for use in subsonic flight generally includes a shaft assembly rotating within a static structure of the gas turbine engine. The bearings supporting the shaft assembly within the static structure must accommodate axial displacement of the shaft assembly due to large changes in temperature and may be subject to a leakage airflow. The gas turbine engine directs a flow of lubricant at the bearings, a portion of which penetrates the bearings and enters the bearing cavity. In some applications, a leakage airflow may impinge directly on the bearing during certain engine operational segments detrimentally affecting lubrication of the bearing.

(3) It would be desirable to have a bearing assembly that accommodates axial movement of a rotating shaft relative to a static structure, one that permits improved lubrication, and one that protects the bearing from airflow.

SUMMARY

(4) According to an aspect of the present disclosure, a gas turbine engine is provided that includes a static structure, a rotating shaft, and a bearing assembly. The rotating shaft is disposed radially inward of the static structure. The bearing assembly includes a bearing and a seal subassembly. The bearing has inner and outer races and a plurality of roller elements disposed between the inner and outer races. The inner race is coupled with the rotating shaft and the outer race is coupled with the static structure. The outer race includes a shield member that extends radially inward toward the rotating shaft. The seal subassembly includes a seal element coupled with the static structure, and a seal runner coupled with the rotating shaft. The seal element is disposed to engage with the seal runner. A portion of the seal runner, a portion of the outer race including the shield member, and a portion of the inner race define a first bearing cavity. A portion of the seal runner, a portion of the outer race including the shield member, a portion of the seal element, and a portion of the static structure define a lateral bearing cavity. The lateral bearing cavity is disposed axially between the first bearing cavity and the seal element. The outer race includes a plurality of drain holes that provide fluid communication between the first bearing cavity and the lateral bearing cavity.

(5) In any of the aspects or embodiments described above and herein, the outer race may have an inner radial surface, an outer radial surface, a first axial end, and a second axial end, and the shield member may extend radially inwardly from the inner radial surface and the shield member may be disposed at the first axial end of the outer race.

(6) In any of the aspects or embodiments described above and herein, the shield member may be integral with the outer race.

(7) In any of the aspects or embodiments described above and herein, each drain hole may extend along an axis disposed at an angle skewed relative to an axial centerline of the engine.

(8) In any of the aspects or embodiments described above and herein, each drain hole may have a first end in communication with the first bearing cavity and a second end in communication with the lateral bearing cavity, and the first end may be disposed radially inward of and axially spaced apart from the second end.

(9) In any of the aspects or embodiments described above and herein, the first end of each drain hole may be disposed at an intersection between the inner radial surface of the outer race and the shield member.

(10) In any of the aspects or embodiments described above and herein, the seal runner may have a curved surface disposed axially between a seal surface and a protrusion that extends outwardly relative to the seal surface.

(11) In any of the aspects or embodiments described above and herein, a portion of the curved surface may form a recess into the seal runner and a portion of the curved surface may extend radially outward to blend with the protrusion.

(12) In any of the aspects or embodiments described above and herein, the seal element may include a seal element housing and a seal ring.

(13) According to an aspect of the present disclosure, a gas turbine engine is provided that includes a static structure, a rotating shaft, and a bearing assembly. The rotating shaft is disposed radially inward of the static structure. The bearing assembly includes a bearing and a seal subassembly. The bearing has inner and outer races and a plurality of roller elements disposed between the inner and outer races. The inner race is coupled with the rotating shaft and the outer race is coupled with the static structure. The outer race includes an inner radial surface, an outer radial surface, and a shield member that extends radially away from the inner radial surface in a direction toward the rotating shaft. The seal subassembly includes a seal element coupled with the static structure, and a seal runner coupled with the rotating shaft. The seal runner has a seal surface and a protrusion that extends outwardly relative to the seal surface. The seal element is disposed to engage with the seal surface of the seal runner. The rotating shaft and the static structure are configured to permit relative axial travel therebetween between a first axial position wherein the seal runner protrusion and the shield member are aligned with one another, and a second axial position wherein the seal runner protrusion and the shield member are misaligned with one another.

(14) In any of the aspects or embodiments described above and herein, a portion of the seal runner, a portion of the outer race including the shield member, and a portion of the inner race may define a first bearing cavity, and a portion of the seal runner, a portion of the outer race including the shield member, a portion of the seal element, and a portion of the static structure may define a lateral bearing cavity. The lateral bearing cavity may be disposed axially between the first bearing cavity and the seal element. The outer race may include a plurality of drain holes that provide fluid communication between the first bearing cavity and the lateral bearing cavity.

(15) In any of the aspects or embodiments described above and herein, the seal element may include a seal element housing and a seal ring, and in the first axial position, the seal ring may engage with a first segment of the seal runner seal surface, and in the second axial position the seal ring may engage with a second segment of the seal runner seal surface.

(16) According to an aspect of the present disclosure, a bearing assembly for use between a static structure and a rotating shaft of a gas turbine engine is provided. The rotating shaft is disposed radially inward of the static structure. The bearing assembly includes a bearing and a seal subassembly. The bearing has inner and outer races and a plurality of roller elements disposed between the inner and outer races. The inner race is configured for coupling with the rotating shaft and the outer race is configured for coupling with the static structure. The outer race has inner and outer radial surfaces, a shield member that extends radially away from the inner radial surface, and a plurality of drain holes. The seal subassembly includes a seal element configured for coupling with the static structure, and a seal runner configured for coupling with the rotating shaft. In an assembled form, the seal subassembly is configured such that the seal element is disposed to engage with the seal runner. The bearing assembly is configured to have an assembled form relative to the static structure and the rotating shaft, and in the assembled form a portion of the seal runner, a portion of the outer race including the shield member, and a portion of the inner race define a first bearing cavity, and in the assembled form a portion of the seal runner, a portion of the outer race including the shield member, a portion of the seal element, and a portion of the static structure define a lateral bearing cavity. The lateral bearing cavity is disposed axially between the first bearing cavity and the seal element. In the assembled form the plurality of drain holes provide fluid

communication between the first bearing cavity and the lateral bearing cavity.

(17) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated otherwise. For example, aspects and/or embodiments of the present disclosure may include any one or more of the individual features or elements disclosed above and/or below alone or in any combination thereof. These features and elements as well as the operation thereof will become more apparent in light of the following description and the accompanying drawings. It should be understood, however, the following description and drawings are intended to be exemplary in nature and non-limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic cross-sectional view of a turbofan gas turbine engine having a bearing arrangement.

(2) FIG. 2 is a cross-sectional view of the casing of FIG. 1 illustrating a bearing assembly in a first axial position.

(3) FIG. 3 is a cross-sectional view of the casing and bearing assembly of FIG. 2 in a second axial position.

DETAILED DESCRIPTION

(4) FIG. 1 illustrates a gas turbine engine 20 configured as a turbofan engine. The gas turbine engine 20 includes a fan section 22, a compressor section 24, a combustor 26, and a turbine section 28 extending along an axial centerline 30. The terms “forward” and “aft” are used herein to indicate position along the axial centerline 30; referring to the engine 20 shown in FIG. 1, the fan section 22 is disposed forward of the compressor section 24, and the compressor section 24 is disposed aft of the fan section 22. The fan section 22 and a portion of the turbine section 28 (i.e., the low pressure turbine) are connected by a low pressure shaft 42. The term “radial” refers to a direction that is perpendicular to the axial centerline 30 and may be used herein to indicate position relative to the axial centerline 30; e.g., a first component positioned “radially inward” of a second component is disposed closer to the axial centerline 30 than the second component, and conversely the second component is disposed “radially outward” of the first component. The term “axial” refers to a direction that is parallel to the axial centerline 30.

(5) The compressor section 24 includes a plurality of compressor rotor stages 32, each stage 32 having a plurality of compressor rotor blades 34 extending radially outward from a hub. The compressor rotor stages 32 and a portion of the turbine section 28 (i.e., the high pressure turbine) are connected by a high pressure shaft 44. The radially outermost surface of a compressor rotor blade 34 may be referred to as the rotor blade tip 36. Each compressor rotor stage 32 is configured for rotation about the axial centerline 30. The compressor section 24 further includes one or more compressor stator vane stages 38. Each compressor stator vane stage 38 includes a plurality of stator vanes 40 (e.g., airfoils) spaced apart from one another around the circumference of the compressor stator vane stage 38. Each compressor stator vane stage 38 is non-rotational about the axial centerline 30. Stator vanes 40 disposed immediately forward of a compressor rotor stage 32 are typically configured to guide airflow into the compressor rotor stage 32 in a desirable orientation. Stator vanes 40 disposed immediately aft of a compressor rotor stage 32 are typically configured to guide airflow exiting the compressor rotor stage 32 in a desirable orientation for downstream use.

(6) Referring still to FIG. 1, the gas turbine engine 20 includes a plurality of bearing assemblies 50 configured to support the low pressure shaft 42 and the high pressure shaft 44. The low pressure shaft 42 is supported at its forward end by a first bearing assembly 50A and a second bearing assembly 50B, and at an aft end thereof by a third bearing assembly 50C. The first bearing

assembly **50A** is disposed axially forward of the second bearing assembly **50B**. The high pressure shaft **44** is supported at its forward end by a fourth bearing assembly **50D** and at an aft end thereof by a fifth bearing assembly **50E**. As will be detailed herein, each bearing assembly **50A-E** includes a bearing suitable for supporting the aforesaid shafts **42**, **44**. Each bearing may be configured to accommodate a radial load and/or a thrust load. Other than as described herein, the present disclosure is not limited to any particular type of bearing. The arrangement of bearing assemblies **50A-E** within the gas turbine engine **20** shown in FIG. **1** is provided as an example only and the present disclosure is not limited thereto.

(7) Referring to FIG. **2**, the bearing assembly **50** discussed in detail below is an example of a present disclosure bearing assembly **50** that may be used between a static structure **52** and a rotating shaft **54**. An engine centerline **30** is shown above the bearing assembly **50** to indicate the orientation of the bearing assembly **50** as shown in FIG. **2**. As will be detailed herein, the bearing assembly **50** includes a bearing **56** and a seal subassembly **80**.

(8) The bearing **56** includes an inner race **60**, an outer race **62**, and a plurality of rolling elements **64** retained therebetween. The outer race **62** includes an inner radial surface **68**, an outer radial surface **65**, a first axial end **67**, a second axial end **69**, and a shield member **66**. The shield member **66** extends radially inwardly from the inner radial surface **68** and is disposed at the first axial end **67** of the outer race **62**. The shield member **66** extends radially inwardly a distance X from the inner radial surface **68** of the outer race **62**. In FIG. **2**, the shield member **66** is shown as integral with the outer race **62**. Alternatively, the shield member **66** may be an independent component that is affixed to the annular outer race **62**. The outer race **62** further includes a plurality of drain holes **70**. Each drain hole **70** extends between a first end **70A** disposed at the inner radial surface **68** and a second end **70B** disposed at the first axial end **67** of the outer race **62**. The plurality of drain holes **70** may be disposed at respective positions spaced apart from one another around the circumference of the outer race **62**. The plurality of drain holes **70** may be equidistantly spaced apart from one another around the circumference of the outer race **62**, but the present disclosure is not limited to equidistant spacing. Each drain hole **70** may extend along an axis **71** that is non-parallel to the engine axial centerline **30**. In the example diagrammatically shown in FIG. **2**, the axis **71** of each drain hole **70** is disposed at an angle α ("a") that is skewed relative to the axial centerline **30**, such that the first end **70A** of each drain hole **70** is disposed radially inward of (and axially separated from) the second end **70B** of the respective drain hole **70**. The angle α (" α ") may be between 40 and 60 degrees. The outer race **62** may have an undercut/chamfer **72** extending between the outer radial surface **65** of the outer race **62** and the first axial end **67** of the outer race **62**. The inner race **60** is affixed to a rotating shaft **54** (e.g., the low pressure shaft **42**) and the outer race **62** is affixed to a static structure **52** (e.g., a static structure of the compressor section **24**). In the example bearing assembly **50** shown in FIG. **2**, the bearing **56** is positioned aft and downstream of the seal subassembly **80**.

(9) The seal subassembly **80** includes a seal element **82** and a seal runner **84**. The seal element **82** is disposed radially outside of the seal runner **84**. The seal element **82** includes a seal element housing **83** and a seal ring **85**. The seal element **82** may be configured to allow the seal ring **85** to float within the seal element housing **83**. The seal element **82** may be a unitary annular structure or may comprise a plurality of elements that collectively form the respective annular structure; e.g., a plurality of circumferential segments. An example of an acceptable seal element **82** is one having a carbon seal ring **85**, but the present disclosure is not limited thereto.

(10) The seal runner **84** may be a unitary annular structure or may comprise a plurality of elements that collectively form the respective annular structure; e.g., a plurality of circumferential segments. The seal runner **84** may include a seal surface **93**, a curved surface **96**, and a radially extending protrusion **98**. The seal surface **93** extends axially and is configured for engagement with the seal ring **85**. As will be described herein, the curved surface **96** is configured to directionally alter a flow of air passing by the curved surface **96**, and the radially extending protrusion **98** may be

configured to impede an airflow passing thereby. The curved surface **96** is disposed axially between the seal surface **93** and the protrusion **98**; i.e., the seal surface **93** axially terminates at the curved surface **96**. In the embodiment shown in FIG. 2, a portion of the curved surface **96** extends radially inward to form a recess into the body of the seal runner, and a portion of the curved surface **96** extends radially outward to blend with the protrusion **98**.

(11) In the bearing assembly **50** diagrammatically shown in FIG. 2, the seal element **82** is connected to the static structure **52** and the seal runner **84** is connected to the rotating shaft **54**. The seal element **82** is configured to permit the seal ring **85** to engage with the seal surface **93** of the seal runner **84**. The engagement between the seal ring **85** and the seal surface **93** is such that a flow of air may flow therebetween during operational segments of the gas turbine engine. Portions of the seal runner **84**, the bearing inner race **60**, the bearing outer race **62**, and the rolling elements **64** define a bearing cavity **90**. Portions of the static structure **52**, the bearing outer race **62**, the seal element **82**, and the seal runner **84** define a lateral bearing cavity **48**.

(12) During operation of the engine **20** and bearing assembly **50**, a lubricant flow **92** is directed at the aft side of the bearing **56**. At least a portion of the lubricant flow **92** will migrate across the bearing **56** and will enter the bearing cavity **90**. At the same time, a difference in static pressure will be present across the seal element **82**; i.e., the static pressure (“P.sub.46”) in the region denoted “46” (on the side of the seal element **82** axially opposite the lateral bearing cavity **48**) will be greater than the static pressure (“P.sub.48”) in the lateral bearing cavity **48** and the static pressure (“P.sub.90”) within the bearing cavity **90**; see FIG. 3. The difference in air pressure across the seal element **82** causes a flow of air (“leakage airflow”) to pass between the seal ring **85** and the seal runner **84** during operational segments of the gas turbine engine **20**. The characteristics (i.e., volumetric rate of flow, air velocity) of the leakage airflow will depend both on the magnitude of the pressure difference across the seal element **82** as well as the size of the gap between the seal ring **85** and the seal surface **93** of the seal runner **84**. Thermal gradients within the engine during operational segments may cause the temperature of rotating shaft **54** to differ from that of the static structure **52**, and thermal expansion of the components may cause the gap between the seal element **82** and the seal surface **93** of the seal runner **84** to vary in magnitude. The gap between the seal ring **85** and the annular seal surface **93** may be circumferentially uniform, or it may vary at different circumferential locations.

(13) The leakage airflow passing between the seal ring **85** and the seal surface **93** (i.e., traveling in a direction from forward to aft) left unchecked may undesirably impinge on the bearing **56**. The present disclosure bearing assembly **50** provides an improved configuration that mitigates or prevents leakage flow impingement on the bearing **56**. For example, the shield member **66** portion of the bearing outer race **62** extending radially toward the seal runner **84** decreases the annular gap therebetween and is understood to block at least some of the leakage airflow. In addition, the radially extending protrusion **98** extends radially toward the shield member **66** of the bearing outer race **62** and decreases the annular gap therebetween. The protrusion **98** is also understood to block at least some of the leakage airflow traveling axially. Still further, the curved surface **96** is understood to direct leakage airflow away from the seal runner **84** and towards the shield member **66** of the outer race **62** (diagrammatically shown as arrow **86**). In this manner, at least a portion of the leakage airflow is directed away from the bearing cavity **90**. That same portion of leakage airflow directed away from the bearing cavity **90** by the curved surface **96** may also interfere with any leakage airflow traveling axially and thereby impede its axial progress.

(14) The drain holes **70** provide an egress passage for lubrication that may collect within the bearing cavity **90**, allowing that lubricant to pass into the lateral bearing cavity **48** where it can be scavenged and returned into the lubrication system.

(15) The bearing assembly **50** is configured to not only accommodate component radial variances as described above but is also configured to accommodate relative axial travel between the rotating shaft **54** (and elements attached thereto) and the static structure **52** (and components attached

thereto). During operation of the engine 20, the relative axial positions of the static portions and rotating portions of the bearing assembly 50 may vary. FIG. 2 illustrates the bearing assembly 50 in a first axial position configuration and FIG. 3 illustrates the bearing assembly 50 in a second axial position configuration. FIG. 3 diagrammatically illustrates the relative axial movement by the distance “AM”. The differences in axial positions can be seen, for example by the relative axial positions of the seal runner 84 and the outer race 62. The bearing 56 has a bearing centerline 57 that bisects the inner race 60 and the rolling elements 64. The bearing centerline 57 does not change relative to the inner race 60 and the rolling elements 64, but may shift relative to the outer race 62, as can be diagrammatically seen in FIGS. 2 and 3. The bearing assembly components (e.g., the shield member 66, the seal runner protrusion 98, and the like are configured to permit the aforesaid axial travel between the rotating shaft 54 (and elements attached thereto) and the static structure 52 (and components attached thereto) without interference.

(16) While the principles of the disclosure have been described above in connection with specific apparatuses and methods, it is to be clearly understood that this description is made only by way of example and not as limitation on the scope of the disclosure. Specific details are given in the above description to provide a thorough understanding of the embodiments. However, it is understood that the embodiments may be practiced without these specific details.

(17) It is noted that the embodiments may be described as a process which is depicted as a flowchart, a flow diagram, a block diagram, etc. Although any one of these structures may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc.

(18) The singular forms “a,” “an,” and “the” refer to one or more than one, unless the context clearly dictates otherwise. For example, the term “comprising a specimen” includes single or plural specimens and is considered equivalent to the phrase “comprising at least one specimen.” The term “or” refers to a single element of stated alternative elements or a combination of two or more elements unless the context clearly indicates otherwise. As used herein, “comprises” means “includes.” Thus, “comprising A or B,” means “including A or B, or A and B,” without excluding additional elements.

(19) It is noted that various connections are set forth between elements in the present description and drawings (the contents of which are included in this disclosure by way of reference). It is noted that these connections are general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect. Any reference to attached, fixed, connected or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option.

(20) No element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprise”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(21) While various inventive aspects, concepts and features of the disclosures may be described and illustrated herein as embodied in combination in the exemplary embodiments, these various aspects, concepts, and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein all such combinations and sub-combinations are intended to be within the scope of the present application. Still further, while various alternative embodiments as to the various aspects, concepts, and features of the disclosures—such as alternative materials, structures, configurations, methods,

devices, and components, and so on—may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed. Those skilled in the art may readily adopt one or more of the inventive aspects, concepts, or features into additional embodiments and uses within the scope of the present application even if such embodiments are not expressly disclosed herein. For example, in the exemplary embodiments described above within the Detailed Description portion of the present specification, elements may be described as individual units and shown as independent of one another to facilitate the description. In alternative embodiments, such elements may be configured as combined elements. It is further noted that various method or process steps for embodiments of the present disclosure are described herein. The description may present method and/or process steps as a particular sequence. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the description should not be construed as a limitation.

Claims

1. A gas turbine engine, comprising: a static structure; a rotating shaft disposed radially inward of the static structure; and a bearing assembly that includes: a bearing having an inner race, an outer race, and a plurality of roller elements disposed between the inner race and the outer race, wherein the inner race is coupled with the rotating shaft and the outer race is coupled with the static structure, and the outer race includes a shield member that extends radially inward toward the rotating shaft; and a seal subassembly that includes a seal element coupled with the static structure, and a seal runner coupled with the rotating shaft, wherein the seal element is disposed to engage with the seal runner; wherein a portion of the seal runner, a portion of the outer race including the shield member, and a portion of the inner race define a first bearing cavity; and wherein a portion of the seal runner, a portion of the outer race including the shield member, a portion of the seal element, and a portion of the static structure define a lateral bearing cavity, and wherein the lateral bearing cavity is disposed axially between the first bearing cavity and the seal element; and wherein the outer race includes a plurality of drain holes that provide fluid communication between the first bearing cavity and the lateral bearing cavity.
2. The gas turbine engine of claim 1, wherein the outer race has an inner radial surface, an outer radial surface, a first axial end, and a second axial end; and wherein the shield member extends radially inwardly from the inner radial surface and the shield member is disposed at the first axial end of the outer race.
3. The gas turbine engine of claim 2, wherein the shield member is integral with the outer race.
4. The gas turbine engine of claim 2, wherein each drain hole of the plurality of drain holes extends along an axis disposed at an angle skewed relative to an axial centerline of the engine.
5. The gas turbine engine of claim 4, wherein each drain hole has a first end in communication with the first bearing cavity and a second end in communication with the lateral bearing cavity, and the first end is disposed radially inward of and axially spaced apart from the second end.
6. The gas turbine engine of claim 5, wherein the first end of each drain hole is disposed at an intersection between the inner radial surface of the outer race and the shield member.
7. The gas turbine engine of claim 1, wherein the seal runner has a curved surface disposed axially between a seal surface and a protrusion that extends outwardly relative to the seal surface.
8. The gas turbine engine of claim 7, wherein a portion of the curved surface forms a recess into the seal runner and a portion of the curved surface extends radially outward to blend with the protrusion.
9. The gas turbine engine of claim 7, wherein the seal element includes a seal element housing and

a seal ring.

10. A gas turbine engine, comprising: a static structure; a rotating shaft disposed radially inward of the static structure; and a bearing assembly that includes: a bearing having an inner race, an outer race, and a plurality of roller elements disposed between the inner race and the outer race, wherein the inner race is coupled with the rotating shaft and the outer race is coupled with the static structure, and the outer race includes an inner radial surface, an outer radial surface, and a shield member that extends radially away from the inner radial surface in a direction toward the rotating shaft; and a seal subassembly that includes a seal element coupled with the static structure, and a seal runner coupled with the rotating shaft, the seal runner having a seal surface and a protrusion that extends outwardly relative to the seal surface; wherein the seal element is disposed to engage with the seal surface of the seal runner; wherein the rotating shaft and the static structure are configured to permit relative axial travel between a first axial position wherein the seal runner protrusion and the shield member are aligned with one another, and a second axial position wherein the seal runner protrusion and the shield member are misaligned with one another.

11. The gas turbine engine of claim 10, wherein a portion of the seal runner, a portion of the outer race including the shield member, and a portion of the inner race define a first bearing cavity; and wherein a portion of the seal runner, a portion of the outer race including the shield member, a portion of the seal element, and a portion of the static structure define a lateral bearing cavity, and wherein the lateral bearing cavity is disposed axially between the first bearing cavity and the seal element; and wherein the outer race includes a plurality of drain holes that provide fluid communication between the first bearing cavity and the lateral bearing cavity.

12. The gas turbine engine of claim 11, wherein each said drain hole of the plurality of drain holes extends along an axis disposed at an angle skewed relative to an axial centerline of the engine.

13. The gas turbine engine of claim 12, wherein a first end of each drain hole is disposed at an intersection between the inner radial surface of the outer race and the shield member.

14. The gas turbine engine of claim 10, wherein the seal runner has a curved surface disposed axially between the seal surface and the seal runner protrusion.

15. The gas turbine engine of claim 14, wherein a portion of the curved surface forms a recess into the seal runner and a portion of the curved surface extends radially outward to blend with the protrusion.

16. The gas turbine engine of claim 10, wherein the seal element includes a seal element housing and a seal ring; and wherein in the first axial position, the seal ring engages with a first segment of the seal runner seal surface, and in the second axial position the seal ring engages with a second segment of the seal runner seal surface.

17. A bearing assembly for use between a static structure and a rotating shaft of a gas turbine engine, wherein the rotating shaft is disposed radially inward of the static structure, the bearing assembly comprising: a bearing having an inner race, an outer race, and a plurality of roller elements disposed between the inner race and the outer race, wherein the inner race is configured for coupling with the rotating shaft and the outer race is configured for coupling with the static structure, and the outer race has an inner radial surface, an outer radial surface, a shield member that extends radially away from the inner radial surface, and a plurality of drain holes; and a seal subassembly that includes a seal element configured for coupling with the static structure, and a seal runner configured for coupling with the rotating shaft; wherein in an assembled form, the seal subassembly is configured such that the seal element is disposed to engage with the seal runner; and wherein the bearing assembly is configured to have an assembled form relative to the static structure and the rotating shaft, and in the assembled form a portion of the seal runner, a portion of the outer race including the shield member, and a portion of the inner race define a first bearing cavity, and in the assembled form a portion of the seal runner, a portion of the outer race including the shield member, a portion of the seal element, and a portion of the static structure define a lateral bearing cavity, the lateral bearing cavity disposed axially between the first bearing cavity and the

seal element, and in the assembled form the plurality of drain holes provide fluid communication between the first bearing cavity and the lateral bearing cavity.

18. The bearing assembly of claim 17, wherein the seal runner has a seal surface and a protrusion that extends outwardly relative to the seal surface, and the seal element is disposed to engage with the seal surface of the seal runner.

19. The bearing assembly of claim 18, wherein the seal runner further includes a curved surface disposed axially between the seal surface and the protrusion, and a portion of the curved surface forms a recess into the seal runner and a portion of the curved surface extends radially outward to blend with the protrusion.

20. The bearing assembly of claim 19, wherein the seal element includes a seal element housing and a seal ring.
